Marmara Üniversitesi
Mühendislik Fakültesi
Elektrik-Elektronik Mühendisliği

Laboratory Manual

EE 3011
Electronics I

Experiment (1)
Diode Rectifier with Peak Detector

Spring 2023

Experiment (1)
Diode Rectifier with Peak Detector

Objectives

- To study the half-wave rectifier function.
- To analyze the output of a peak detector with regulator.

Pre-lab Work

Analyze the circuit below using the constant voltage diode model. The input is a sinusoidal wave with amplitude V_m and frequency f. Find V_o with and without the capacitor C. In each case, draw the waveform of the output and find the average output voltage and average load current.

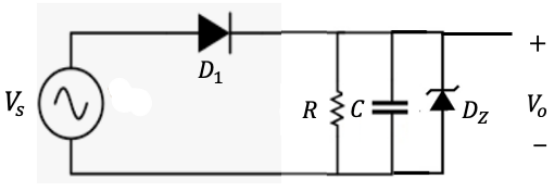


Figure 1: Half-wave rectifier with peak detector and regulator.

Solution

Equipment

Table 1. Equipment for experiment 5.

	Equipment	Model	Serial No.	Stock No.
1	BreadBoard Station			
2	1 kΩ Resistor (2W, 5%)			
3	Capacitors (2)			
4	Diodes (2)	1N4001 (1) 1N4732 (1)		
5	Multi-meter (1)			
6	Dual-trace Oscilloscope			
7	Probes and Cables			

Experimental Procedure

Part (1): Half-wave Rectifier Output Analysis.

1. Connect the circuit in figure 1 without the capacitor and the Zener diode and with R=1 kΩ. Pay a special attention to how you connect the terminals of the diode. The input voltage source V_S is sinusoidal.
2. Connect channel 1 of the oscilloscope to V_S and adjust it to 10 V amplitude at 60 Hz. Pay attention to the polarity of the oscilloscope connection. Sketch the input waveform in graph 1 labeling all important points in both voltage and time axes. Make your sketch as accurate as possible.

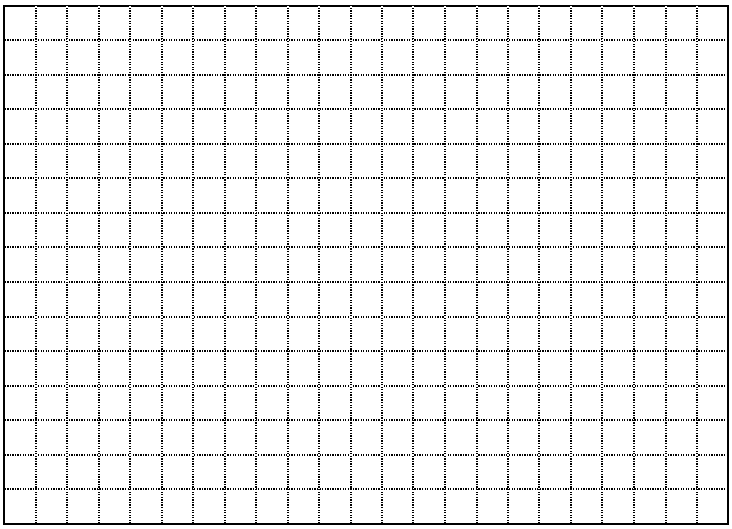
3. Now connect channel 2 of the oscilloscope to V_o. Sketch the output waveform on the same graph 1 labeling all important points in both voltage and time axes. Make your sketch as accurate as possible. Use different colors for the waveforms.
4. Using the multi-meter, measure the output voltage and record it in table 2.
5. Repeat steps 1-4 for a triangular V_s waveform. Use graph 2 for the sketches.

Part (2): Half-wave Rectifier with Peak Detector and Regulator.

1. Consider again the circuit in figure 1 with the capacitor connected only.
2. Repeat the whole procedure in part 1 for both capacitors provided. Record results in graphs 1 and 2 and table 3. Use different colors for the waveforms.
3. Finally, connect the Zener diode across the capacitor and observe the output. Record results in graphs 1 and 2. Use different colors for the waveforms.

Results and Analysis

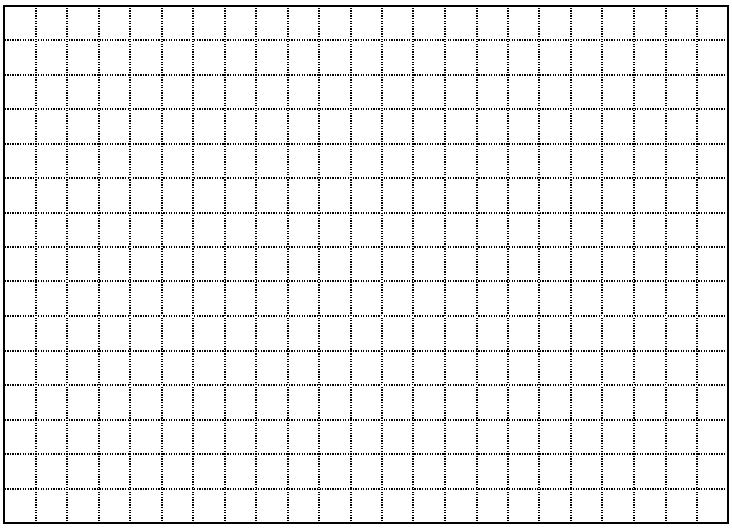
Part (1): Half-wave Rectifier Output Analysis.



Graph 1: Sinusoidal input case.

Table 2. Average output voltage of the half-wave rectifier without peak detector.

Sinusoidal input			Triangular input		
V_o (ave) (V) Experimental value	V_o (ave) (V) Theoretical value	% Difference	V_o (ave) (V) Experimental value	V_o (ave) (V) Theoretical value	% Difference



Graph 2: Triangular input case.

1. Calculate the theoretical value of the average output voltage for both input waveform types and record it in table 2.
2. Comment on your results regarding the waveforms in graphs 1 and 2.

3. Comment on your results regarding the average output voltage.

Part (2): Half-wave Rectifier with Peak Detector and Regulator.

Table 3. Average output voltage of the half-wave rectifier with peak detector.

	Sinusoidal input			Triangular input		
Capacitor value (μF)	$V_{O\,(ave)}$ (V) Experimental value	$V_{O\,(ave)}$ (V) Theoretical value	% Difference	$V_{O\,(ave)}$ (V) Experimental value	$V_{O\,(ave)}$ (V) Theoretical value	% Difference

1. Calculate the theoretical value of the average output voltage for both capacitor values and record it in table 3.
2. Comment on your results regarding the waveforms in graphs 1 and 2.

3. Comment on your results regarding the average output voltage

LTspice Simulations

Include with your report LTspice simulation results of all parts of this experiment.

Conclusions and Remarks

Write below all your conclusions/remarks on this experiment.

Name		Name		Name		Name	
ID		ID		ID		ID	
Section							